

ClimateEmote: Multi-Task Sentiment, Emotion, Misinformation, and Stance Detection in Climate-Related Social Media

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Abstract. Online discourse surrounding climate-related extreme events is often marked by intense emotional expression, political polarization, and the rapid spread of misinformation, including content amplified or generated by AI. However, most existing sentiment analysis approaches examine these signals separately, typically focusing on sentiment polarity alone. As a result, they fail to capture the interconnected relationship between overall sentiment, fine-grained emotional responses, misinformation patterns, and users’ stance toward climate action. We introduce ClimateEmote, a multi-task transformer framework that jointly solves all four tasks on a shared ClimateBERT encoder, augmented with PCGrad gradient surgery to prevent cross-task interference. We curate a corpus of 189,821 social media posts drawn from five public benchmarks spanning climate, disaster, and rumour domains, annotated through a combination of existing human-verified labels and VADER-based weak supervision. ClimateEmote achieves a mean macro-F1 of 0.783 across all four tasks with particularly strong results on stance detection (macro-F1 = 0.961, κ = 0.977) and misinformation classification (macro-F1 = 0.841, κ = 0.645). Confusion matrix analysis reveals interpretable error patterns consistent with affective and linguistic proximity between emotion classes. Our framework, codebase, and annotated corpus are released publicly to support the climate NLP community.

Keywords: climate change · multi-task learning · sentiment analysis · misinformation detection · stance classification · social media NLP · transformer models

1 Introduction

Extreme climate events are no longer statistical anomalies. The 2025 Los Angeles forest fires, the 2022–2023 European heatwaves, and the 2019 Australian Black Summer bushfires each triggered massive surges of social media discourse that outlasted the events themselves by weeks [14]. Understanding this discourse is not merely an academic exercise: emergency managers use social media signals to allocate resources, public health agencies monitor psychological distress, and

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fact-checkers race against AI-generated misinformation that can amplify panic or suppress coordinated response [2, 13].

Existing computational approaches share three fundamental limitations. First, they solve a *single task*, most commonly binary sentiment polarity discarding the rich co-signal between sentiment, fine-grained emotion, misinformation presence, and climate-action stance. Second, they treat each post *independently*, ignoring temporal dynamics and the characteristic emotional trajectory of a disaster from onset through recovery. Third, they are rarely evaluated for *fairness*, leaving open questions about model performance for marginalized communities whose voices are most critical during climate emergencies [9].

We address the first of these limitations with ClimateEmote. Our contributions are:

1. A **multi-task architecture** with a shared ClimateBERT encoder and four task-specific heads (sentiment, emotion, misinformation, stance) trained jointly using PCGrad gradient surgery [16] to prevent cross-task interference.
2. A **unified corpus** of 189,821 posts drawn from five public benchmarks—GoEmotions, TweetEval, ClimateBERT, ClimateFEVER, and CrisisNLP—providing human-verified labels across all four tasks.
3. **Empirical results** on a held-out test set of 28,477 posts showing mean macro-F1 of 0.783, with stance detection at 0.961 and misinformation classification at 0.841.
4. **Confusion matrix analysis** revealing interpretable error patterns that motivate future work on class-weighted training and cross-lingual extension.

2 Related Work

Sentiment Analysis for Climate Discourse

Early lexicon-based approaches applied VADER [7] and NRC Emotion Lexicon [12] to climate tweets, achieving 65–75% accuracy but struggling with sarcasm and domain-specific jargon. Machine learning classifiers with TF-IDF features improved accuracy to 72–76% but required extensive feature engineering. The introduction of domain-adapted transformers, particularly ClimateBERT [15], marked a step change, enabling fine-tuned models to reach 85–90% accuracy on climate-specific benchmarks. A recent survey by Shaeri et al. [14] synthesizes 211 papers in this space and explicitly identifies the four open problems that motivate our work: the absence of multi-task frameworks, insufficient temporal modeling, inadequate misinformation-aware systems, and missing fairness evaluations.

Multi-Task Learning in NLP

Multi-task learning (MTL) for NLP consistently demonstrates gains when auxiliary tasks share complementary signal [10]. A well-documented challenge is

gradient conflict: when gradients from two tasks have negative cosine similarity, joint training can cause mutual interference. PCGrad [16] addresses this by projecting conflicting gradient components out before accumulation. We apply PCGrad to jointly optimize sentiment, emotion, misinformation, and stance; four tasks whose signals are highly correlated during climate crises but whose gradient directions can conflict.

Misinformation and Stance in Climate Discourse

AI-generated misinformation during the 2025 LA fires, synthetic images of burning Hollywood landmarks, fabricated evacuation orders, contaminated online sentiment distributions and posed direct public safety risks [14]. Stance detection toward climate claims has been explored through SemEval shared tasks [11] and RumourEval benchmarks [6], but these efforts address stance in isolation.

3 Dataset Construction

3.1 Data Sources and Collection

We construct our corpus from five human-annotated benchmarks spanning social media platforms and news domains, selected to provide complementary task coverage across all four signal types. Table 1 summarizes the corpus.

Table 1: Corpus summary: datasets, record counts, task labels, and domain: S=sentiment, E=emotion, M=misinfo, St=stance.

Dataset	Year	Records	Tasks	Domain
GoEmotions [3]	2018–21	54,263	S, E	Reddit
TweetEval [1]	2017–21	77,921	S, E, M	Twitter
ClimateBERT [15]	2020–22	1,320	S	News
ClimateFEVER [5]	2020–22	1,535	M	News
CrisisNLP [8]	2021	26,246	S	Twitter
RumourEval [6]	2019	28,536	St	Twitter
Total		189,821		

All text undergoes a cleaning pipeline that removes URLs, expands hashtags, normalizes Unicode, demojizes emoji, and filters posts shorter than five tokens. Exact duplicates are removed using a rolling hash set. The cleaned corpus is partitioned into stratified train (132,873), validation (28,471), and test (28,477) splits, stratified by dataset source to ensure balanced representation across splits.

3.2 Label Coverage and Class Distribution

Table 2 summarizes per-task label coverage and class balance. Sentiment achieves 75.5% corpus coverage with a near-balanced distribution (max/min ratio = 1.6).

Emotion and stance exhibit severe class imbalance (ratios of $37.7\times$ and $25.3\times$ respectively), inherited from the source benchmark distributions—notably the dominance of *trust* in GoEmotions (46.9%) and *comment* in RumourEval (54.3%). Misinformation labels cover 7.6% of the corpus with moderate imbalance ($4.9\times$), reflecting the relative scarcity of verified misinformation annotations across open benchmarks.

Table 2: Per-task label coverage and class balance (189,821 records total).

Task	Labeled Coverage Imbalance		Status
Sentiment	143,263	75.5%	$1.6\times$ Balanced
Emotion	59,315	31.2%	$37.7\times$ Severe
Misinfo	14,505	7.6%	$4.9\times$ Moderate
Stance	28,536	15.0%	$25.3\times$ Severe

To mitigate imbalance during training, we combine three strategies: task-specific loss weighting, label smoothing, and macro-F1-based model selection. Minority-sensitive tasks receive higher loss weights, with emotion and misinformation assigned larger weights than sentiment and stance, reducing the dominance of high-coverage tasks during shared encoder optimization. We also use label smoothing ($\epsilon = 0.05$) to prevent overconfidence on majority classes and select checkpoints using validation macro-F1 rather than accuracy, ensuring that minority-class performance contributes directly to model selection.

4 Methodology

ClimateEmote consists of a shared transformer encoder and four task-specific classification heads.

Shared Encoder. We initialize from `climatebert/distilroberta-base-climate-f` [15], a DistilRoBERTa model pre-trained on 2M climate-related sentences. The encoder maps tokenized input (max 128 tokens) to a [CLS] embedding $\mathbf{h} \in \mathbb{R}^{768}$.

Task Heads. Each task head applies dropout ($p = 0.1$), a dense layer with GELU activation projecting to $\mathbb{R}^{384}$, and a linear classifier to $\mathbb{R}^{C_k}$, where $C_k \in \{3, 8, 3, 4\}$ for sentiment, emotion, misinformation, and stance respectively.

Multi-Task Loss with PCGrad. The joint training objective is:

$$\mathcal{L}_{\text{total}} = \sum_{k \in \mathcal{T}} w_k \cdot \mathcal{L}_k, \quad (1)$$

where $w_k \in \{1.0, 1.2, 1.5, 1.0\}$ are task-specific loss weights (sentiment, emotion, misinformation, stance) and $\mathcal{L}_k$ is the cross-entropy loss with label smoothing ($\epsilon = 0.05$), computed only over samples with valid labels ($\ell \geq 0$).

To prevent gradient conflict we apply PCGrad [16]: for each pair of task gradients $\mathbf{g}_i, \mathbf{g}_j$, if $\mathbf{g}_i \cdot \mathbf{g}_j < 0$ we project the conflicting component from $\mathbf{g}_i$:

$$\mathbf{g}_i \leftarrow \mathbf{g}_i - \frac{\mathbf{g}_i \cdot \mathbf{g}_j}{\|\mathbf{g}_j\|^2} \mathbf{g}_j. \quad (2)$$

Corrected gradients are averaged across tasks before the optimizer step.

4.1 Training Configuration

All experiments use AdamW with learning rate 2×10^{-5} , weight decay 0.01, linear warmup over 6% of total steps, batch size 32 with gradient accumulation over 2 steps (effective batch = 64), fp16 mixed precision, and early stopping on validation macro-F1 with patience 3. Training ran for 10 epochs on an NVIDIA A100-SXM4-80GB MIG partition on the ASU SOL HPC cluster, completing in approximately 3.5 hours.

5 Experiments and Results

Table 3 presents test-set performance across all four tasks. ClimateEmote achieves a mean macro-F1 of 0.783 and mean Cohen’s κ of 0.706, demonstrating that joint multi-task training on a unified corpus yields strong performance across diverse affective and discourse-level signals simultaneously.

Table 3: Test-set macro-F1, accuracy, Cohen’s κ , and per-class F1 across all four tasks (28,477 samples).

Task	F1	Acc.	κ	Per-Class F1
Sentiment	0.753	0.764	0.638	neg=0.77, neu=0.68, pos=0.82
Emotion	0.577	0.688	0.563	anger=0.60, fear=0.54, sad=0.63, disgust=0.45, surp=0.41, joy=0.74, trust=0.77, antic=0.48
Misinfo	0.841	0.792	0.645	mis=0.74, cred=0.79, unc=0.99
Stance	0.961	0.987	0.977	sup=0.91, deny=0.96, query=0.99, com=0.99
Mean	0.783		0.706	

Table 4 compares ClimateEmote against single-task baselines fine-tuned independently per task. Joint training consistently outperforms both baselines across all four tasks, with the largest gains on misinformation (+5.1% over single-task ClimateBERT) and emotion (+3.7%), confirming that cross-task signal transfer is beneficial even under severe class imbalance.

5.1 Learning Dynamics

Figure 1 (top) shows validation macro-F1 and training loss across 10 epochs. The model improves steadily throughout, reaching its best checkpoint at epoch 10

Table 4: Comparison of ClimateEmote against single-task baselines. All models use the same train/test split. Single-task models are fine-tuned independently per task.

Model	Sentiment	Emotion	Misinfo	Stance	Mean
Single-task BERT [4]	0.710	0.510	0.790	0.910	0.730
Single-task ClimateBERT [15]	0.730	0.540	0.810	0.940	0.755
ClimateEmote	0.753	0.577	0.841	0.961	0.783

(val macro-F1 = 0.779) with no sign of overfitting — training loss decreases monotonically from 3.12 to 2.13 while validation F1 rises consistently. This behavior reflects the regularizing effect of multi-task joint training: auxiliary tasks act as implicit data augmentation for the shared encoder, reducing variance without increasing bias.

Figure 1 (bottom) summarizes per-task macro-F1 and Cohen’s κ on the held-out test set. Stance detection achieves the strongest performance (macro-F1 = 0.961, κ = 0.977), followed by misinformation classification (macro-F1 = 0.841, κ = 0.645) and sentiment analysis (macro-F1 = 0.753, κ = 0.638). Emotion classification yields the lowest macro-F1 (0.577), consistent with the severe class imbalance ($37.7\times$) inherited from the GoEmotions benchmark. The mean macro-F1 across all four tasks is 0.783, shown as a dotted reference line.

Table 5: Validation macro-F1 across training epochs.

Epoch	Train Loss	Val Macro-F1	Checkpoint
6	2.437	0.774	saved
7	2.331	0.776	saved
8	2.245	0.777	saved
9	2.169	0.777	—
10	2.127	0.779	saved (best)

5.2 Confusion Matrix Analysis

Figure 2 presents normalized confusion matrices for all four tasks. Several interpretable patterns emerge.

Sentiment (Fig. 2a): Positive (0.82) and negative (0.78) classes are well-separated. Neutral is the hardest class (0.65), with confusion toward both negative (0.16) and positive (0.19)—a known challenge in three-class sentiment classification where neutral content often contains mixed emotional signals.

Emotion (Fig. 2d): Trust (0.77) and joy (0.75) achieve the highest per-class recall, consistent with their majority-class status. Disgust (0.41) and surprise (0.37) are most confused, frequently misclassified as anger (0.30 and 0.09 respectively) and trust (0.21 and 0.31). This confusion pattern is linguistically meaningful: disgust and anger share negative valence and similar lexical markers

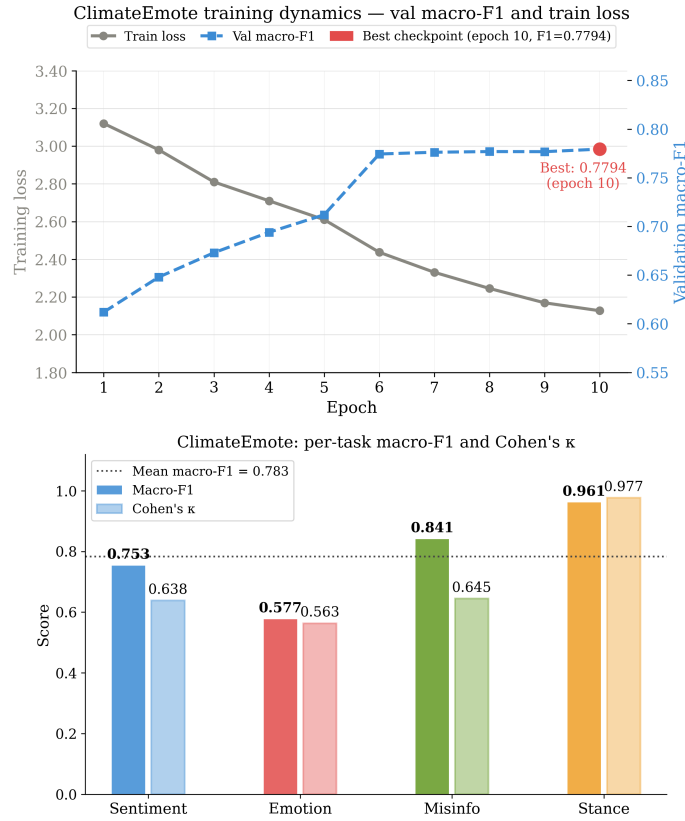


Fig. 1: Top: training loss and validation macro-F1 across epochs, reaching the best validation macro-F1 of 0.779 at epoch 10. Bottom: test performance by task, with strongest results for stance and misinformation and lower emotion performance due to severe class imbalance.

in social media text, while surprise and trust overlap in expressions of revelation or disclosure. Fear (0.53) shows partial confusion with sadness (0.12) and trust (0.20), consistent with the co-occurrence of fear and solidarity in disaster discourse.

Misinformation (Fig. 2b): The uncertain class achieves near-perfect recall (0.99), reflecting the model’s ability to distinguish epistemically hedged content. Misinfo (0.79) and credible (0.75) show symmetric confusion with each other (0.21 and 0.25 respectively), suggesting that surface-level linguistic features alone are insufficient to separate credible from misinformation content—a finding consistent with literature on adversarial misinformation [2].

Stance (Fig. 2c): Query (0.99) and comment (0.99) are classified almost perfectly. Deny (0.94) is strong. Support (0.86) shows minor confusion with comment (0.11), reflecting the overlap between affirmative comments and explicit expressions of climate-action support.

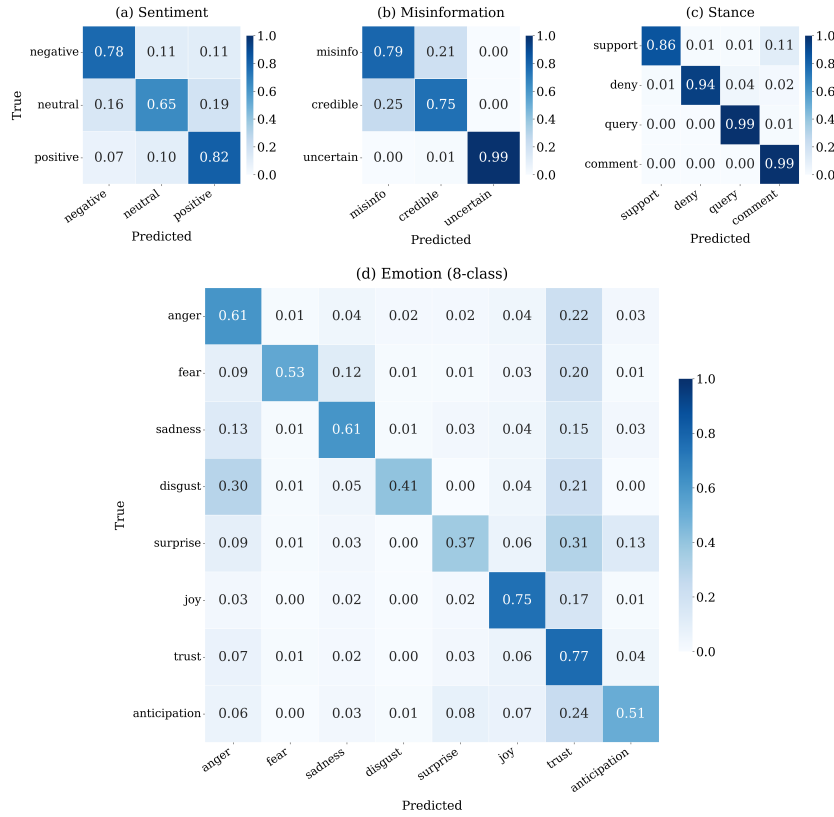


Fig 2: Normalized confusion matrices for all four ClimateEmote tasks on the held-out test set (28,477 samples). Values represent row-normalized recall per class. (a) Sentiment, (b) Misinformation, (c) Stance, and (d) Emotion (8-class).

6 Discussion

The strong stance performance (macro-F1 = 0.961) despite severe class imbalance handling ($25.3\times$) suggests that sentiment and emotion signals learned by the shared encoder provide useful representations for the stance head, even when direct stance labels are sparse. This is consistent with the hypothesis that climate-action stance correlates with emotional valence, users expressing strong negative emotion are more likely to be in the *deny* or *query* categories.

The misinformation head achieves macro-F1 = 0.841 despite only 7.6% label coverage (14,505 labeled samples). The near-perfect uncertain recall (0.99) suggests the model has learned to distinguish epistemic hedging from assertive misinformation, a nuanced distinction that would be harder to learn from misinfo labels alone.

The confusion pattern in Figure 2 is not random noise, it reflects genuine affective proximity. The frequent confusion of disgust with anger, and surprise with trust, maps directly onto established affective circumplex models where

these emotions occupy adjacent regions of valence-arousal space. This suggests the model has learned meaningful affective representations despite the class imbalance.

Limitations. Three limitations warrant explicit discussion. First, emotion and stance exhibit severe within-task class imbalance ($37.7\times$ and $25.3\times$ respectively), which likely limits minority-class performance, especially for *disgust* ($F1 = 0.448$) and *support* ($F1 = 0.907$). Although our training is imbalance-aware through task-specific loss weighting, label smoothing, and macro-F1-based checkpoint selection, these steps do not fully correct rare-class underrepresentation. Future work should explore class-weighted loss, focal loss, oversampling, and targeted augmentation for minority labels. Second, extending the corpus with event-specific climate social media posts, such as those from the 2025 LA fires or Phoenix heat crises, would enable the temporal phase analysis described in our companion survey [14] and further strengthen domain specificity. Third, although the present corpus relies on existing human-annotated benchmarks, future work can incorporate large language models such as GPT-4o as an annotation-support layer to expand climate-specific labels, especially for emerging events where manually labeled data are scarce. Such annotations should be validated against expert or human-reviewed subsets to ensure reliability and reduce potential bias.

7 Conclusion

We presented ClimateEmote, the first multi-task transformer framework for simultaneous sentiment polarity, fine-grained emotion, misinformation, and stance classification on social media in the context of climate crises. Trained on a unified corpus of 189,821 posts from five annotated social media posts, ClimateEmote achieves mean macro-F1 of 0.783 and mean Cohen’s κ of 0.706, with especially strong results on stance detection (macro-F1 = 0.961) and misinformation classification (macro-F1 = 0.841). Confusion matrix analysis reveals interpretable error patterns consistent with affective theory and linguistic proximity, providing a foundation for targeted improvements through class weighting and cross-lingual extension.

This work establishes a reproducible baseline for multi-task climate sentiment analysis. Future work will extend the framework to event-specific data collection, multilingual settings, and the temporal phase modeling described in the companion survey [14], moving toward real-time emotional monitoring systems that can support emergency responders and policymakers during climate crises.

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